

MODULE 2: Data Warehousing

INTRODUCTION TO DATA WAREHOUSING

A data warehouse is a centralized repository designed for storing, organizing, and analyzing large volumes of structured data from various sources. It serves as the backbone of business intelligence (BI) and decision-making processes by enabling efficient querying, analysis and reporting. The following are some of the key considerations in designing and managing a data warehouse

Data Warehousing Architecture

The architecture of a data warehouse typically includes the following components:

Data sources

operational systems are the primary sources of data and include transactional databases like sales, CRM and ERP; web servers; and sensors. Data can also be sourced from external sources like social media, weather data, and market research.

ETL Processes

These are automated procedures to extract, transform, and Load (ETL) data from the data sources. They extract the data, transform it to be suitable for use, and then load or transfer it to the data warehouse storage. The extraction process makes use of techniques such as database connections, APIs and file transfers. In the transformation process, Data is cleaned, formatted, and integrated into a consistent format suitable for analysis. Data cleaning involves identification and correction of errors, and data validation to ensure quality standards are met. Formatting will include converting of fields such as dates to a consistent format. Data integration deals with combining data from different sources to create a unified record about one entity of interest. Another consideration in the ETL process is the refresh intervals. Since data in the data warehouse is historical in nature, it needs to be regularly refreshed via the ETL process so as it can include the most current data.

Data Storage Database

This is the central repository where the integrated and cleansed data is stored. The database is optimized for analytical queries and reporting, often using schemas like star or snowflake (discussed later) to organize the data effectively.

Access Tools

These tools enable users to interact with the data warehouse and retrieve information for analysis. They include reporting tools, OLAP (Online Analytical Processing) tools, and business intelligence applications that facilitate data mining, querying and visualization of the data. Examples of such tools are Tableau, Power BI and custom-built dashboards.

Metadata Management

Metadata provides context and information about the data itself. It is data about the data offering a guide about its structure, origin, and meaning. Effective metadata management is critical for the success of any data warehouse. Organizations need to establish and maintain a centralized repository for storing and managing metadata. This repository should be easily accessible to data analysts, data scientists, and other data users. Additionally, regular updates and maintenance of metadata are essential to ensure its accuracy and relevance.

Metadata can be broadly categorized into several types:

Technical Metadata

This type of metadata describes the technical characteristics of the data, such as:

1. Data Source - Origin of the data for example, operational system, external file).
2. Data Format - File type for example, CSV, XML,JSON), data encoding, and data types for example, integer, string, date).
3. Storage Location -Physical location of the data within the data warehouse for example, database tables, file paths).
4. Data Volume and Size - Information about the size and volume of data.
5. Data Lineage - Tracking the history and transformations of data from source to destination.

Business Metadata

This type of metadata provides context and meaning to the data from a business perspective. It includes:

1. Business Glossary - Definitions of key business terms and concepts.
2. Data Definitions - Descriptions of data fields and their meanings.
3. Business Rules - Constraints and validation rules for data.

4. Data Ownership - Assigning responsibility for data quality and accuracy to specific individuals or teams.
5. Data Usage - Information about how the data is used for business analysis and decision-making.

Operational Metadata

This type of metadata focuses on the operational aspects of data management, such as:

1. ETL Process Metadata - Information about the ETL processes used to extract, transform, and load data into the data warehouse.
2. Job Schedules - Schedules for data extraction, transformation, and loading tasks.
3. User Access Controls - Information about user permissions and access rights to data.
4. Data Quality Rules - These are rules and procedures for ensuring data accuracy, completeness, and consistency.

3. Data Mart Architecture

A data mart, within the broader context of data warehousing, serves as a specialized repository of data tailored to the specific analytical needs of a particular business unit such as a department. Unlike a data warehouse, which aims to integrate data from across the entire organization, a data mart focuses on a specific subject area, such as sales, marketing, finance, or human resources. It contains data relevant to the needs of a particular department. This focused approach offers several key advantages. Firstly, it simplifies data analysis by reducing the volume of data that needs to be processed and analyzed. By focusing on a specific subject area, data marts make it easier for business users to understand and utilize the data effectively. Secondly, data marts can often provide faster query performance compared to a larger, more complex data warehouse, enabling quicker insights and more agile decisionmaking.

Data marts can be categorized as either dependent or independent. Dependent data marts extract data directly from a larger, enterprise-wide data warehouse. In contrast, independent data marts source data directly from various operational systems within the organization.

DATA WAREHOUSE MODELS

In a data warehouse, data is usually represented in the form of a data cube. A data cube is a way to represent the several attributes of an item of interest or

fact. An item of interest can for instances be a single sale of one item. The attributes of interest can include the customer, the product, the customer, the date and time of sale and the sales person. These attributes are called dimensions. Each dimension may also have attributes. For instance, the customer may have attribute such as name, gender, age, phone, town, county, and country. The cube metaphor is therefore a way to represent an event of interest along multiple dimensions.

Dimensions

Dimensions are fundamental building blocks and provide context and meaning to the core data, known as facts. They act as lenses through which analysts can view and analyze the facts, enabling them to gain deeper insights and answer complex business questions.

You can consider dimensions as the "who," "what," "where," "when," and "why" of your data. The who is the customer dimension (customer ID, name, address, demographics); the what is the product dimension (product ID, name, category, brand); the where is the Location dimension (store ID, city, state, country); the when is the time dimension (date, time, year, quarter); the why is the reason dimension (promotion ID, campaign name).

The following are some key characteristics of dimensions

1. i.Descriptive - Dimensions are primarily composed of descriptive attributes that provide context and meaning to the facts.
2. Hierarchical - Many dimensions exhibit hierarchical structures. For example, the time dimension can be hierarchically organized into years, quarters, months, and days.
3. Static or Slowly Changing - Some dimensions, such as customer demographics, may change slowly over time, while others, such as product information, may change more frequently.

You use dimensions to represent your data in the following way:

1. Fact Tables - These tables store the core measurements or "facts" about the business, such as sales, revenue, costs, or website traffic.
2. Dimension Tables -These tables provide detailed information about the dimensions associated with the facts.
3. Relationships - Fact tables are linked to dimension tables through foreign key relationships, allowing analysts to analyze facts within the context of different dimensions.

Database Schemas

There are two data representation models that are used to represent data in a data warehouse. These are the star schema and the snowflake schema.

The Star Schema

The star schema is one of the simplest and most widely used database schemas for data warehousing, particularly in the context of Online Analytical Processing (OLAP). It organizes data into fact tables and dimension tables, forming a structure that resembles a star when diagrammed. The central "fact" table holds

numerical data (measures), and it is connected to various dimension tables that contain descriptive attributes related to the facts.

Star Schema Example

Let's consider a Sales Data Warehouse for a retail company. The star schema would look like this:

A. Fact Table

Sales Fact Table

The Sales Fact Table stores the key metrics of the sales process, such as sales revenue, units sold, and discounts applied. Each row in the fact table represents a specific sales transaction, identified by keys from the dimension tables.

Transaction ID	Customer ID	Product † ID	Time ID	Store ID	Revenue	Units Sold	Discount
1	1001	2001	1	101	500	10	0.05
2	1002	2002	1	102	300	5	0.10
3	1003	2003	2	103	700	7	0.02

Dimension Tables

Each dimension table contains descriptive data that allows users to analyze the fact data from different perspectives.

Customer Dimension Table

Customer ID	Name	Age Group	Gender	Town	County
1001	John Kamau	25–34	Male	Naivasha	Nakuru
1002	Jane Auma	35–44	Female	Gilgil	Nakuru
1003	Alice Makena	45–54	Female	Chogoria	Tharaka Nithi

Product Dimension Table

Product ID	Product Name	Category	Brand	Price
2001	Laptop A	Electronics	Brand X	500
2002	Smartphone B	Electronics	Brand Y	300
2003	Chair C	Furniture	Brand Z	100

Time Dimension Table

Time ID	Year	Quarter	Month	Day	Weekday
1	2024	Q1	January	15	Monday
2	2024	Q1	February	10	Saturday

Store Dimension Table

Store ID	Store Name	Location	Manager
101	Store 1	Nakuru	Juma
102	Store 2	Kisumu	Pamela
103	Store 3	Meru	Alima

In this star schema, given any fact, we can get all the required information about it. For instance the transaction with id is for customer John Kamau from Naivasha, Nakuru County. We can also get information about the store, the product, and the time. We can also similarly obtain summary information for instance about all the sales in a given store or a given county.

The following are some of the main benefits of the star schema:

1. **Simplicity** - The star schema is easy to understand and implement. It provides a straightforward structure where the fact table is surrounded by dimension tables, forming a star-like shape.
2. **Optimized for Query Performance** -The denormalization of dimension tables (storing redundant data in a single table) allows for faster query performance compared to normalized structures. The fact table can be easily queried without needing to join multiple normalized tables.
3. **Efficient Data Aggregation** - The schema is designed to efficiently perform aggregation operations like summing, averaging, or counting at different levels of granularity for example, by month, region, or product).
4. **Flexibility in Analysis** -Users can easily "slice" and "dice" the data from various angles. For example, you can quickly analyze sales by product, by time, by region, or by customer segment.

The Star Schema

The snowflake schema is an extension of the star schema with dimensions tables referencing other dimensions tables. It normalizes the dimension tables to reduce data redundancy by breaking them into multiple related tables, forming a structure that resembles a snowflake. While the star schema has denormalized dimensions, the snowflake schema organizes dimensions into a set of related tables, improving data integrity and potentially reducing storage requirements. For this schema, the queries are usually more complex but updates are easier.

Snowflake Schema Example

- Fact Table

Sales Fact Table

The Sales Fact Table remains the same, storing the key metrics of the sales process such as sales revenue, units sold, and discounts applied, along with foreign keys linking to the dimension tables.

Transaction ID	Customer ID	Product ID	Time ID	Store ID	Revenue	Units Sold	Discount
1	1001	2001	1	1	500	10	0.05
2	1002	2002	1	2	300	5	0.10
3	1003	2003	2	4	700	7	0.02

Normalized Dimension Tables

Unlike the star schema, the snowflake schema normalizes the dimension tables into multiple related tables. Here's how we would split the Customer, Product, Time, and Store dimensions into normalized structures.

Customer Dimension

Customer ID	Name	Age Group ID	Gender ID	Town ID
1001	John Kamau	1	1	1
1002	Jane Auma	Alice Makena	3	2
1003				

Age Group Table

Age Group ID	Age Group
1	25 – 34
2	35 – 44
3	45 – 54

Gender Table

Gender ID	Gender
1	Male
2	Female

County Table

County ID	County Name
1	Nakuru
2	Kisumu
3	Tharaka-Nithi
4	Meru

Town Table

Town ID	Town Name	County ID
1	Naivasha	1

2	Gilgil	1
3	Chogoria	3
4	Nkubu	4

Product Dimension

Product ID	Product Name	Category ID	Brand ID
2001	Laptop A	1	1
2002	Smartphone B	1	2
2003	Chair C	2	3

Category Table

Category ID	Category
1	Electronics
2	Furniture

a. Brand Table (separate table for brand):

Brand ID	Brand
1	Brand X
2	Brand Y
3	Brand Z

Time Dimension

Time ID	Year	Quarter ID	Month ID
1	2024	1	1
2	2024	1	2

Quarter Table

Quarter ID	Quarter
1	Q 1
2	Q 2
3	Q 3
4	Q 4

Month Table

Month ID	Month
1	January
2	February

Store Dimension

Store ID	Store Name	Town ID	Manager ID
101	Store 1	1	1
102	Store 2	2	2
103	Store 3	4	3

Manager Table

Manager ID	Manager
1	Mark Taylor
2	Sarah Lee
3	John Smith

Fact constellation

In a fact constellation schema is use in sophisticated applications may require multiple fact tables to share which share the dimensions tables.

”This kind of schema can be viewed as a collection of stars, and hence is called a galaxy schema or a fact constellation (Han, J., & Kamber, M, 2018).